Time-Series Analysis of CS:GO/CS2 Skin Prices: Stationarity, Breaks, and Volatility Forecasting

Essay

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1 Introduction

1.1 Motivation

The subject of the analysis in this paper is the Counter Strike skins market. It is quite recent phenomenon and it is actively developing. In just two years it has grown from 2.2 billion USD to 6 billion USD, and continues to grow. Hence there is almost no research, although it has some distinct features and might be quite useful for asset trading understanding. Nevertheless, there are some papers, even the HSE one.
These papers and the author's own experience in trading those in-game skins were the main source of interest for further analysis. Some of portfolios (called inventories) have vastly outperformed the SnP500 and Cryptocurrencies returns. And the news of growing cybersports industry and game development propel the bloom of the market. Of course, there is a risk-factor in the face of Valve - the game developer, who might influence the market directly, causing structural risks. But it is also an unexplored field that might be used for diversification of investor's portfolios yielding high returns, and thus is subject to the analysis.

1.2 Data source

Unfortunately, aggregate indices of the market are behind the paywall. As this project is not financed by any institution, the author was to choose some reasonable dataset and perform the analysis on this part.
The data is taken from the GitHub. For further analysis only two skins were chosen: m4a4 Desolate Space, Field-Tested and m4a1-s Decimator, Field-Tested. This is mainly due to the interrelation of the two weapons. In the game it is the main weapon of the CT side for all players (basically, available for each player in half of in-game rounds that he plays). They might be seen as substiututes, this is evident from the price movements and usage rate statistics (when one gun is nerfed, the usage rate shifts towards the other and vise versa). Skins are also mid-tier by price - affordable for the majority of players, but valuable enough to be an item of choice - as many cheaper skins are used for crafts, not for actual in-game use. This mechanics allows to obtain more expensive skins from less expensive - thus the price of latter is mainly shaped by the price of mid-tier skins. Also, the quality of the skin - Field-Tested means that this is the most liquid version of the skin, as other qualities' skins are more rare and have less market transactions, making the data more sparse and volatile, which is worse for the purpose of the analysis.

1.3 Main question

The aim of this work is to explore integrated behavior of prices and returns of the above skins and look at predictive power of the forecasts obtained exploring this property.

2 Data

The dataset used in this study consists of daily historical prices for two CS:GO/CS2 weapon skins: M4A4 — Desolate Space (Field-Tested) and M4A1-S — Decimator (Field-Tested). The raw prices were obtained from the Buff.163 marketplace, the second most active trading platform in the CS skin market after the official Steam Community Market. Buff.163 provides reliable and liquid pricing information, especially for frequently traded rifle skins.

The sample covers the period from **26 July 2021** to **9 February 2024**, comprising 826 daily observations for each price series and 825 daily log returns correspondingly. Prices are recorded in Chinese yuan (CNY).

Let $P_{1,t}$ denote the daily price of the M4A1-S Decimator and $P_{2,t}$ the price of the M4A4 Desolate Space. Log prices are defined as $p_{i,t} = \log P_{i,t}$, and daily log returns as $r_{i,t} = p_{i,t} - p_{i,t-1}$ for $i = 1, 2$.

Basic descriptive statistics for the price and return series are reported in Table 1. The price series display slow-moving dynamics with visible level shifts, often caused by game updates, while the return series fluctuate around zero with elevated volatility. Figures 1 and 2 show the evolution of log prices and log returns over the sample period. Autocorrelation functions (ACF) and partial autocorrelation functions (PACF) of the return series (Figures 3) show rapid decorrelation, suggesting that returns exhibit weak serial dependence but nontrivial conditional heteroskedasticity, motivating the use of GARCH-type models for the analysis.

Table 1: Summary Statistics for Prices and Log Returns

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
$P_t^{(1)}$ (Desolate Space)	27.940	7.388	17.05	48.25	0.411	2.160
$P_t^{(2)}$ (Decimator)	59.221	9.470	33.47	82.25	-0.123	2.605
$r_t^{(1)}$ (Return, Desolate)	0.00138	0.0484	-0.173	0.8692	11.871	202.555
$r_t^{(2)}$ (Return, Decimator)	0.00127	0.0226	-0.118	0.10095	-0.105	5.439

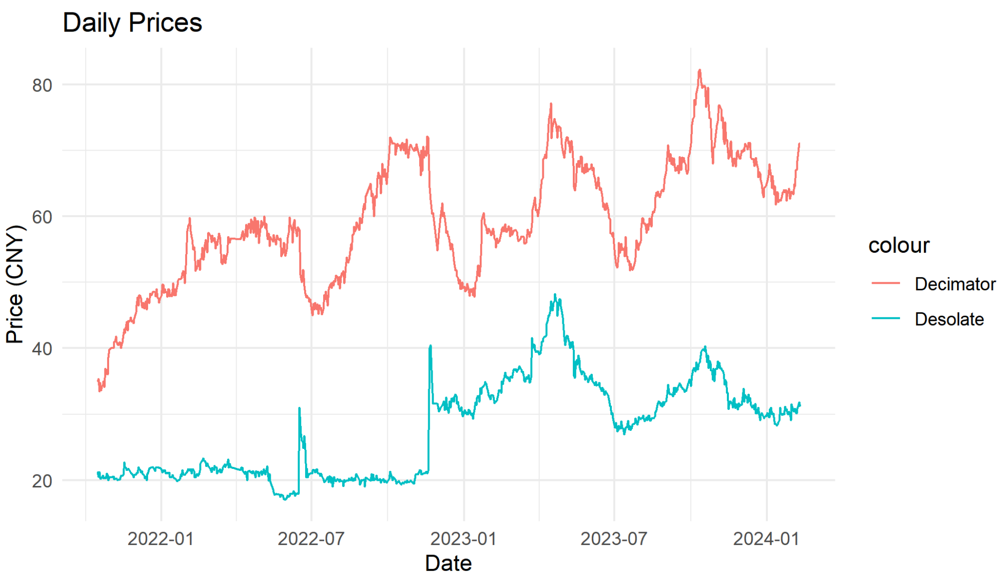


Figure 1: Daily Prices of M4A4 Desolate Space and M4A1-S Decimator (CNY)

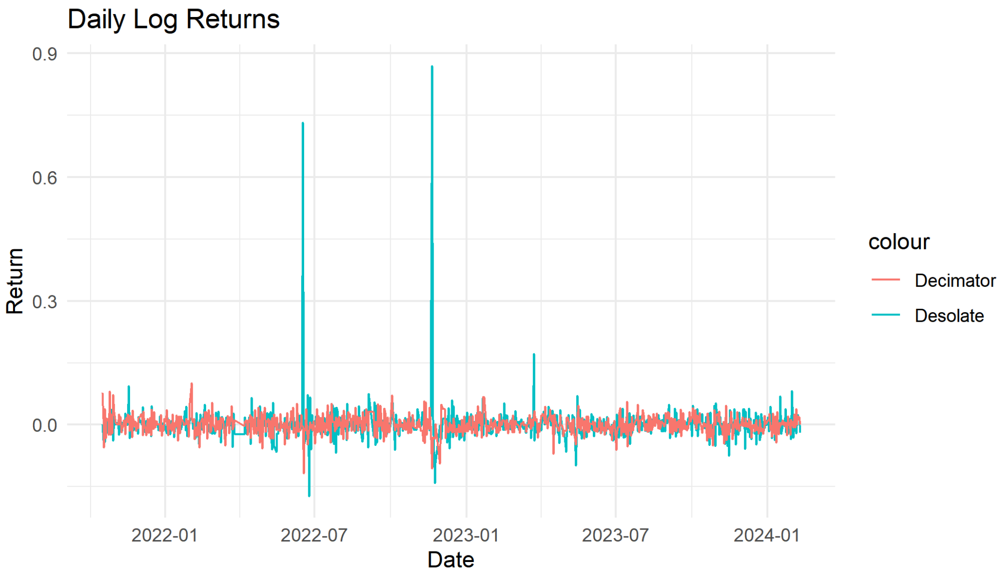
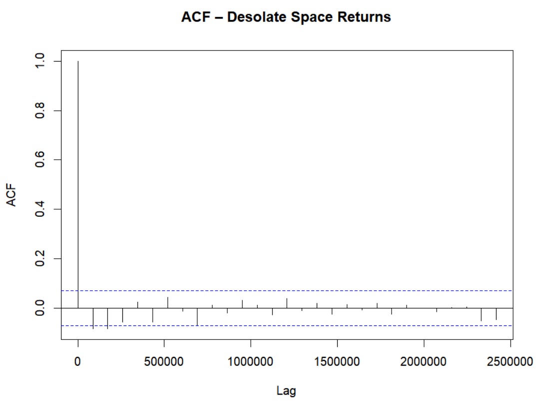
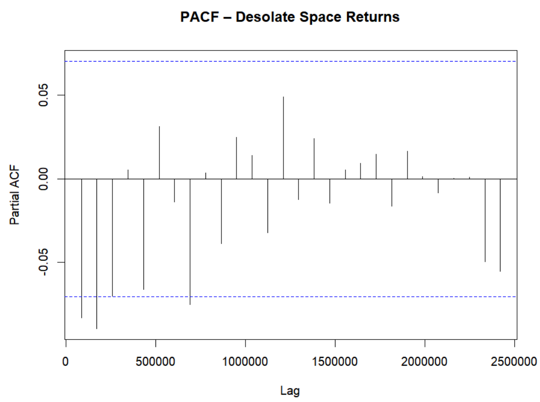


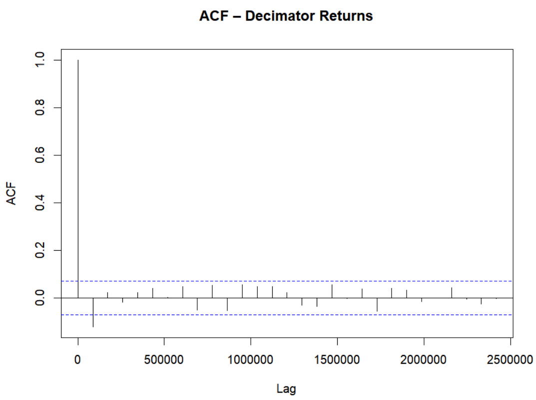
Figure 2: Daily Log Returns of M4A4 Desolate Space and M4A1-S Decimator



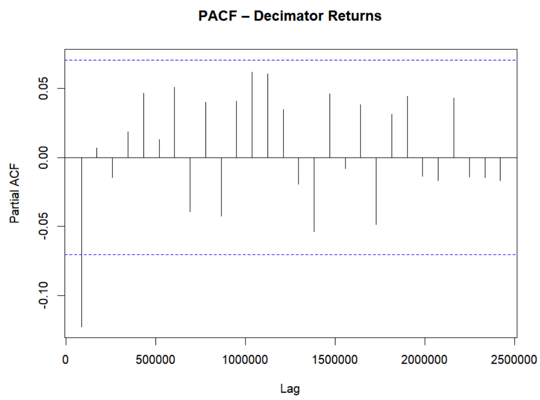
(a) ACF – Desolate Space



(b) PACF – Desolate Space



(c) ACF – Decimator



(d) PACF – Decimator

Figure 3: ACF and PACF Plots for Log Returns

Additionally, it can be seen that there are structural breaks, followed by level shifts. The primary sources of those are game updates introducing those weapons' balancing and major game updates, which attract new audience (like switch from CS:GO to CS2). Returns have very heavy tails (kurtosis > 100) and fluctuate around 0. Simultaneously, prices suffer from autocorrelation and non-stationarity (as will be shown further) and returns show almost no autocorrelation. Concluding, prices suffer from non-stationarity and are affected by the structural breaks, thus subsequent analysis will mainly focus on returns. Latter show stationarity and volatility clustering, which maker them suitable for simple ARMA specifications (ARMA(0,0) for Decimator and ARMA(1,0) for Desolate Space) and GARCH(1, 1).

3 Stationarity Analysis

Table 2: ADF Unit Root Tests

Series	Stat.	5% Crit.	Conclusion
$\log P_t^{(1)}$ (Desolate)	-2.04	-2.86	Non-stationary
$\log P_t^{(2)}$ (Decimator)	-2.46	-2.86	Non-stationary
$r_t^{(1)}$ (Desolate)	-22.32	-1.95	Stationary
$r_t^{(2)}$ (Decimator)	-20.72	-1.95	Stationary

Table 3: KPSS Stationarity Tests

Series	Stat.	5% Crit.	Conclusion
Level null: Stationarity			
$\log P_t^{(1)}$ (Desolate)	7.364	0.463	Non-stationary
$\log P_t^{(2)}$ (Decimator)	5.798	0.463	Non-stationary
$r_t^{(1)}$ (Desolate)	0.073	0.463	Stationary
$r_t^{(2)}$ (Decimator)	0.206	0.463	Stationary
Trend null: Trend-stationarity			
$\log P_t^{(1)}$ (Desolate)	0.964	0.146	Non-stationary
$\log P_t^{(2)}$ (Decimator)	0.343	0.146	Non-stationary
$r_t^{(1)}$ (Desolate)	0.060	0.146	Stationary
$r_t^{(2)}$ (Decimator)	0.078	0.146	Stationary

Both log price series fail the ADF test for stationarity and strongly reject the KPSS null of level and trend stationarity, indicating that prices are non-stationary. This is consistent with the visual evidence of trends and structural breaks associated with major game updates. In contrast, the return series strongly reject the ADF unit root null and fail to reject the KPSS stationarity null, confirming that both return series are stationary. Therefore, all subsequent modeling is performed using returns rather than price levels.

4 Univariate Time Series Models

The analysis in this section is applied to the stationary time series - log returns.

4.1 ARMA Models

The ARMA specifications for returns were selected using the Bayesian Information Criterion (BIC) over the grid ARMA(p, q) with $p, q \in \{0, 1, 2\}$. For the Decimator series, BIC selected an ARMA(0,0) model, corresponding to a white-noise process with zero mean. For the Desolate Space series, BIC selected an AR(1) model. The resulting ARMA models are summarised in Table 4.

Table 4: ARMA Model Selection for Returns

Series	Selected Model	σ^2	BIC
$r_t^{(1)}$ (Decimator)	ARMA(0,0)	0.002339	-2490.28
$r_t^{(2)}$ (Desolate)	ARMA(1,0)	0.0005045	-3676.62

4.2 GARCH Models

Despite white-noise behaviour in the conditional mean, both return series exhibit volatility clustering and heavy tails, motivating conditional heteroskedasticity models. For each return series, a GARCH(1,1) model was estimated using the ARMA orders selected above. Parameter estimates are shown in Tables 5 and 6.

Table 5: GARCH(1,1) Estimates for Decimator Returns

Parameter	Estimate	Robust SE	t-value
μ	0.000272	0.000000	5563.4***
ω	0.000098	0.000000	8457.6***
α	0.167533	0.000019	8596.0***
β	0.647492	0.000081	7992.7***
Log-likelihood		1666.08	
AIC		-4.2948	
BIC		-4.2708	

NOTES: All coefficients are highly statistically significant under robust errors.

Table 6: GARCH(1,1) Estimates for Desolate Space Returns

Parameter	Estimate	Robust SE	t-value
μ	0.001030	0.000732	1.41
AR(1)	-0.145835	0.052006	-2.80**
ω	0.000048	0.000027	1.78*
α	0.116559	0.045292	2.57**
β	0.789644	0.082288	9.60***
Log-likelihood		1861.74	
AIC		-4.7978	
BIC		-4.7677	

NOTES: Several coefficients are statistically significant under robust errors.

For the M4A1-S Decimator return series, the estimated GARCH parameters satisfy $\alpha + \beta = 0.82$, implying mean-reverting but persistent volatility. For the M4A4 Desolate Space series, $\alpha + \beta = 0.91$, suggesting even more persistent volatility and slower decay of shocks. Diagnostic tests (Ljung–Box, ARCH LM) confirm that both models adequately capture serial correlation and heteroskedasticity. No significant sign or leverage effects were detected, indicating that symmetric GARCH(1,1) models are suitable for the analysis.

4.3 Explaination

Both time series reflect the properties of mean-reverting volatility as $\alpha + \beta < 1$. No autocorrelation is found with Ljung-Box test. Also, constant-variance model fit has a lower log-likelihood, which makes GARCH an appropriate setup. Some coefficients appear less precisely estimated due to the heavy-tailed return distributions, although several parameters — particularly in the Desolate Space model — remain statistically significant even under robust standard errors. Heavy tails inflate robust standard errors, but diagnostic tests still indicate that the GARCH(1,1) model is appropriate in this case.

4.4 In-sample fit observations

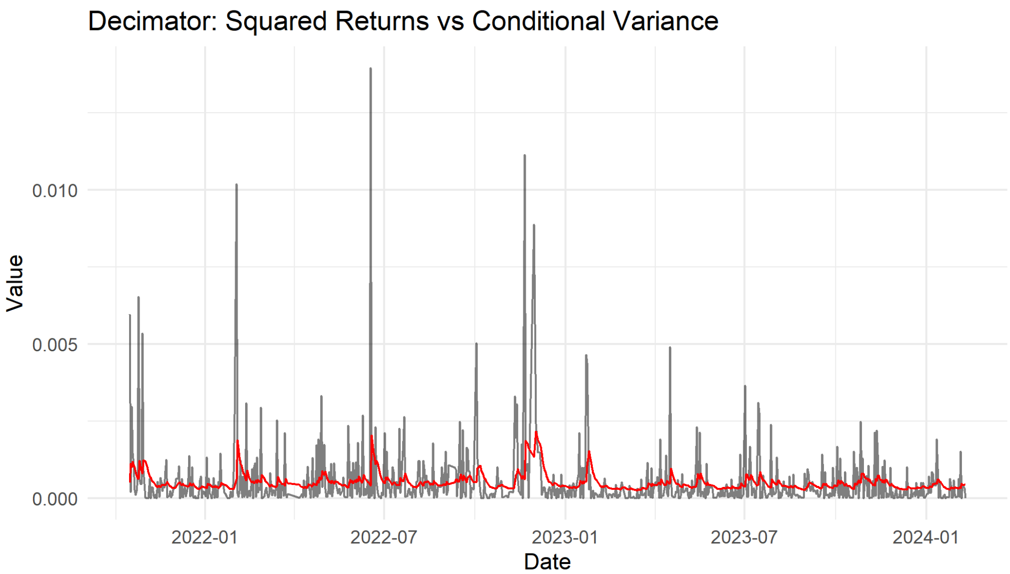


Figure 4: Squared returns (grey) and the GARCH(1,1) one-step-ahead conditional variance forecast (red) for the M4A1-S Decimator.

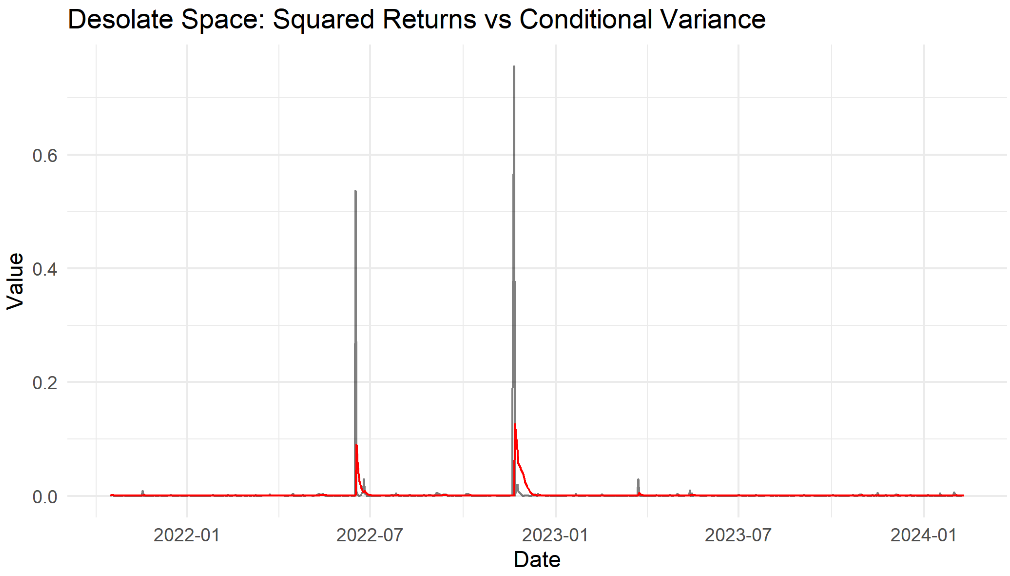


Figure 5: Squared returns (grey) and the GARCH(1,1) one-step-ahead conditional variance forecast (red) for the M4A4 Desolate Space skin.

Evidently, GARCH fit consistently captures the volatility shifts, however it is less volatile, which indicates heavier-tailed distribution of the data.

5 Structural Breaks

To investigate the structural breaks, I use the Bai-Perron test for multiple breakpoints to see wether they coincide with game updates.

Figures 6 and 7 display the estimated breakpoints. The log price series for both skins exhibit clear structural shifts, consistent with market reactions to major game updates that directly affect the in-game usability and demand for the underlying weapon skins.

Table 7 reports the estimated break dates. These structural breaks provide further evidence that the price level is non-stationary, supporting the decision to model return series rather than price levels. Moreover, the presence of breaks highlights that volatility dynamics may differ across regimes, although the return series remained stationary after first-differencing.

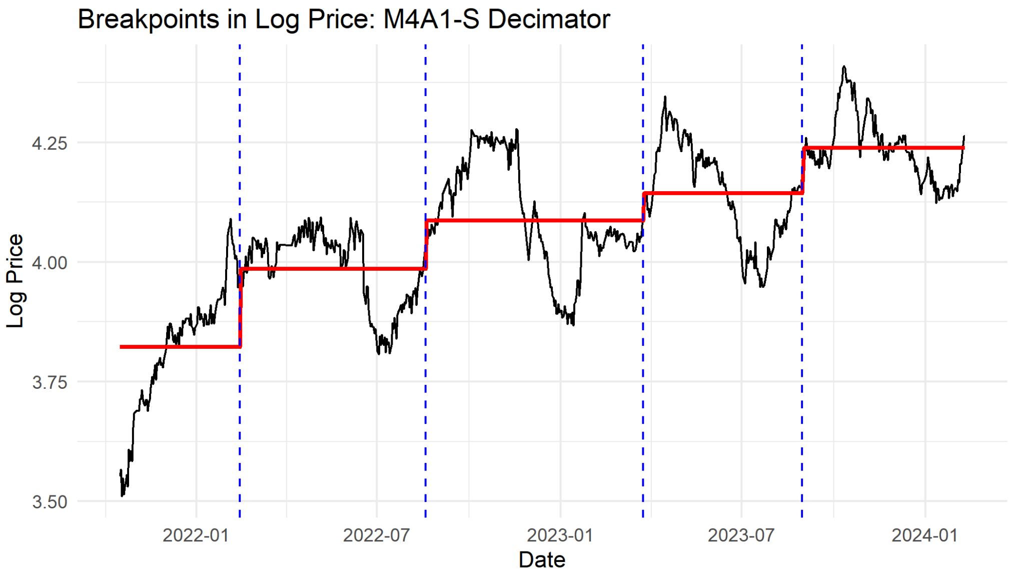


Figure 6: Bai–Perron structural breaks in the log price of M4A1-S Decimator. Black line: log price; red line: fitted piecewise means; blue dashed lines: estimated breakpoints.

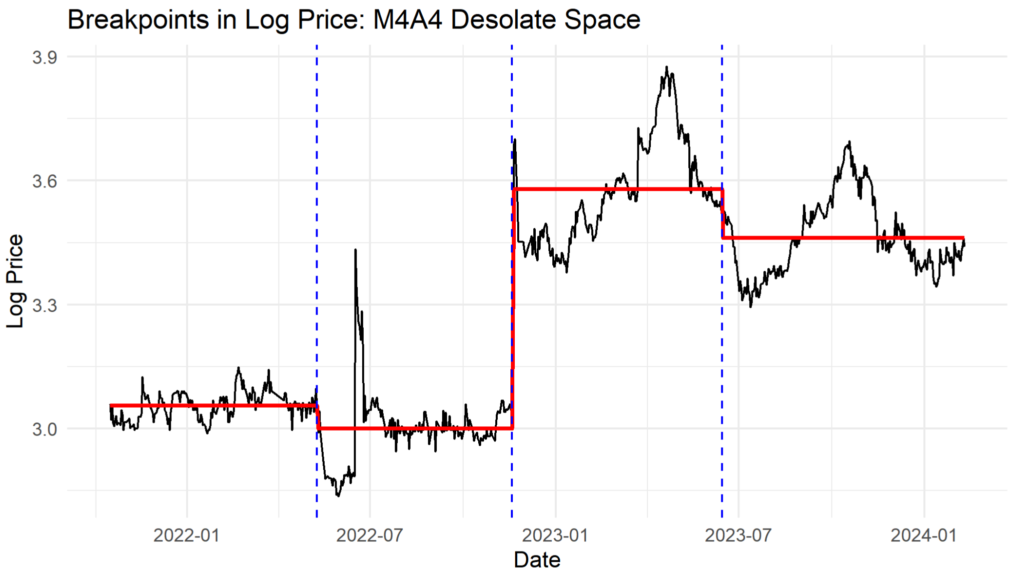


Figure 7: Bai–Perron structural breaks in the log price of M4A4 Desolate Space. Black line: log price; red line: fitted piecewise means; blue dashed lines: estimated breakpoints.

Table 7: Estimated Bai–Perron Break Dates in Log Prices

Series	Breaks	Estimated Break Dates
M4A1-S Decimator	4	13 Feb 2022; 18 Aug 2022; 24 Mar 2023; 30 Aug 2023
M4A4 Desolate Space	3	09 May 2022; 18 Nov 2022; 14 Jun 2023

There were several in-game updates associated with these breaks. First, on 22-nd September 2021 M4A1-S received a significant buff, and as player could choose only one of those two weapons in that version of the game, players' preferences have shifted towards the Decimator and other M4A1-S skins from M4A4 skins, including the Desolate Space. As can be seen from the graph there is a huge gap in prices, some time before the found breakpoint. This is mainly due to the gradual reaction of the market.

Later, on 2022 November the 18-th, the update has nerfed the M4A1-S, and Bai-Perron test finds this date perfectly, as the reaction on Desolate Space lot was instant. The situation on the Decimator graph looks different. At that times there was a clear trend upwards, mainly due to the limited supply of the skin and growing number of players. Notably, there are dips on both graphs, corresponding to the dates of winter and summer sales in Steam, when people are willing to sell their skins and buy games with discounts.

6 Forecasting

To evaluate the out-of-sample performance of the fitted ARMA–GARCH models, a rolling one-step-ahead forecasting scheme was employed. The final 200 observations of each return series were reserved as the evaluation sample, while the remaining history served as the initial estimation window. At each forecast origin, the ARMA–GARCH model was re-estimated using all data available up to that point, and a single-step forecast of both the conditional mean and conditional variance was generated. The window was then expanded by one observation, and the procedure repeated until forecasts were obtained for the entire 200-day out-of-sample period. The results of this procedure are represented in the table 8 and can be seen on figures 8 and 9.

Table 8: Out-of-sample one-step-ahead forecast accuracy for returns and volatility.

Series	RMSE	MAE	QLIKE
M4A1-S Decimator	0.0234	0.0178	-6.4769
M4A4 Desolate Space	0.0185	0.0147	-6.9649

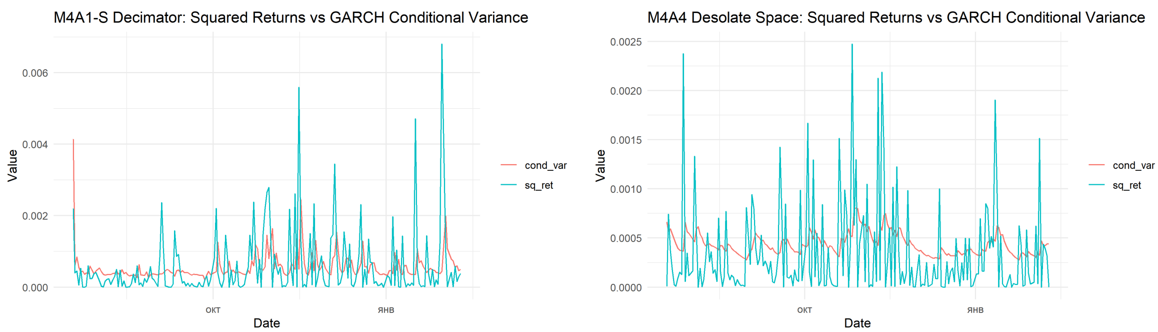


Figure 8: Squared returns and one-step-ahead GARCH(1,1) conditional variance forecasts for the M4A1-S Decimator (left) and M4A4 Desolate Space (right) over the out-of-sample period.

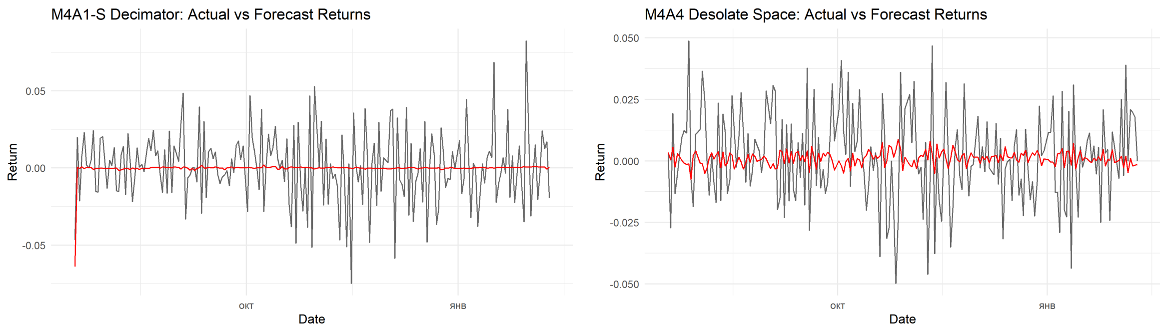


Figure 9: Actual vs one-step-ahead forecast returns for the M4A1-S Decimator (left) and M4A4 Desolate Space (right) over the out-of-sample period.

Thus, ARMA models are describing white noise or weak AR(1) behavior, and therefore are not interesting for the analysis. Whereas GARCH models capture volatility clustering phenomenon, but there is still a room for a better forecast, waiting in the final section.

7 Multivariate Analysis

Exploring the inter-series connectoins it is evident that there exists a seasonal co-movement. Naturally, the idea to estimate joint seasonality was implemented in the following way. Averaging the log prices the composite index was created, later it was decomposed with the help of STL to extract shared seasonality. This common index is reported in figure 10.

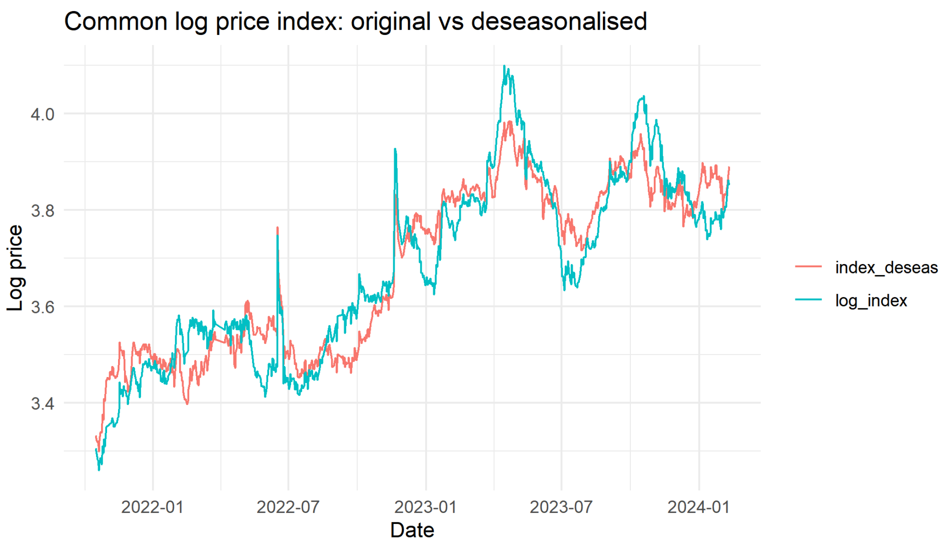


Figure 10: Common log price index and its deseasonalised counterpart obtained from STL decomposition.

Extraction of common seasonal component makes series smoother, especially in the absence of structural breaks, which spoil the work of STL. Nonetheless, series became more skin-specific, as can be seen on figure 11.

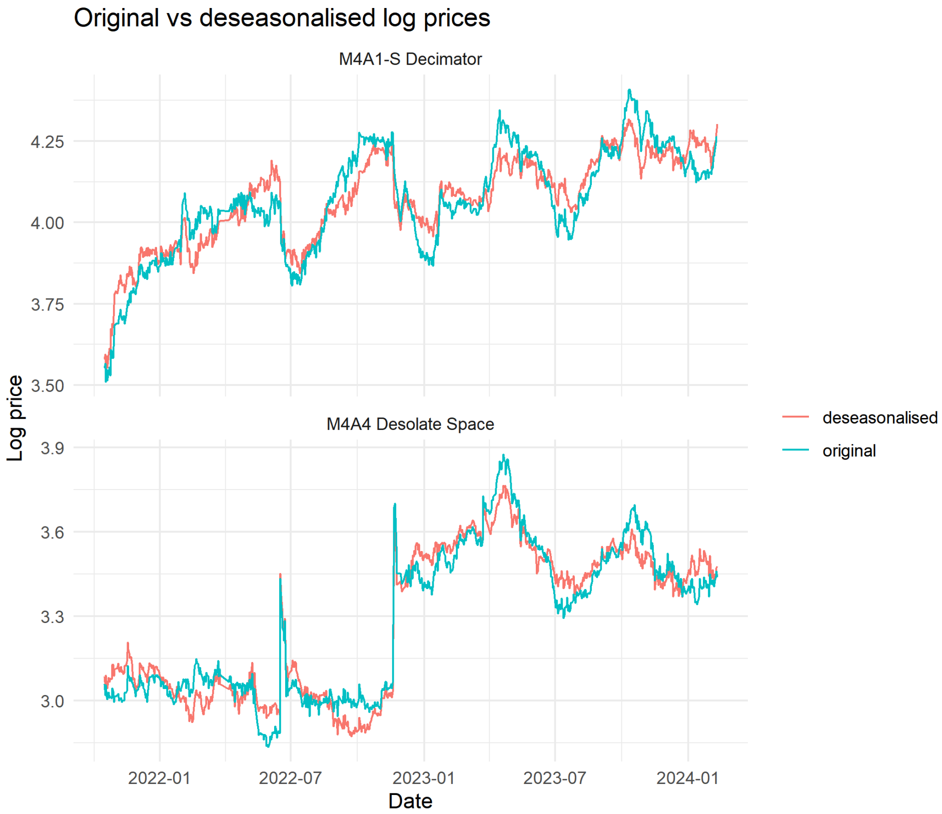


Figure 11: Original and deseasonalised log prices for the M4A1-S Decimator and M4A4 Desolate Space skins.

It turned out that data behavior corresponds to the slow-moving polynomial trend, which can be found in figure 12. More precisely, this data is still to some extent seasonal, but the estimation of this effect is rather hard in presence of structural shocks, as written above. Overall, the multivariate patterns confirm that while both skins respond to common shocks, their medium-horizon trajectories diverge, suggesting that basket-level modelling or VAR-type approaches may be more suitable for capturing their shared structure than purely univariate models.

With more reliable and longer-period data and proper software it is possible to construct a reliable panel dataset out of the liquid skins with daily trading frequency. However, due to current limitations, the author's analysis stops here.

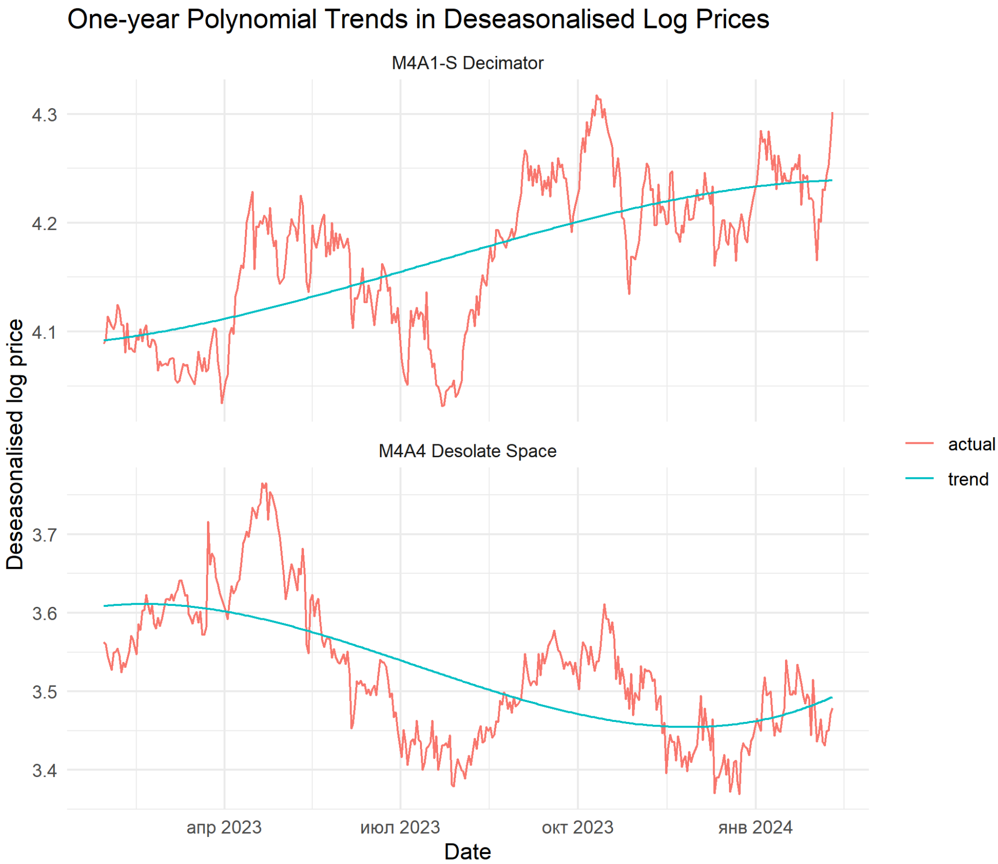


Figure 12: One-year cubic polynomial trends fitted to the deseasonalised log prices of the M4A1-S Decimator and M4A4 Desolate Space skins. The black lines show the deseasonalised series, while the red lines depict the estimated smooth trends over the final 365 days of the sample.

8 Conclusion

This study analysed the time-series properties of two liquid Counter Strike skins using a combination of stationarity tests, ARMA–GARCH modelling, structural break detection and multivariate decomposition methods. Both price series were found to be non-stationary with pronounced structural breaks corresponding to major game updates and market-wide events, while returns were stationary but exhibited heavy tails and volatility clustering. Consequently, ARMA–GARCH models provided an adequate representation of short-run return dynamics, capturing shifts in volatility but offering weak predictive power for return levels, as reflected in the modest out-of-sample performance.

Multivariate analysis has proven that both skins share common cyclical and seasonal patterns, most notably semi-annual dips associated with Steam sales, which was initially expected. Deseasonalisation and polynomial trend extraction made medium-run behaviour more transparent, highlighting divergence in underlying trends despite substantial co-movement and providing more meaningful forecasting basis. These findings demonstrate that digital asset markets such as videogame skins exhibit many of the stylised facts found in traditional financial markets—non-stationarity, structural shifts, volatility persistence—yet also contain domain-specific seasonal effects and event-driven behaviour. Future work may explore regime-switching volatility models, higher-frequency data, or multivariate GARCH specifications to further improve forecasting and risk assessment in this emerging market.

Remarkably, Valve's Steam Subscriber Agreement clarifies that users receive a limited license to game items, revocable at their discretion, without granting property rights or investment value. This aligns with their bans on promoting skin gambling, case-opening, or trading sites at official events, distinguishing these from classical assets. That being said, these items carry many properties of financial assets and may represent the important foundation for other financial innovations analysis.

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